Muhammad Ariff Bin Abdul Razak

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WORK EXPERIENCE

Machine Learning & Fullstack Engineer (Intern), KLASS Engineering and Solutions
Sep 2025 – Aug 2026

- Reproduced and benchmarked state-of-the-art AI-generated video and audio detectors (NSG-VD, D3, ReStraV, Speech-Forensics) on SLURM-based HPC, achieving 90.03 AUROC on the NSG-VD replication.
- Integrated optical flow (RAFT), segmentation (SAM 2), and point-tracking (CoTracker) algorithms to analyze physical motion derivatives in video sequences for mission-critical data.
- Engineered a data generation pipeline chaining video extension (Wan2.2), voice cloning (Chatterbox), and lip-synchronization (LatentSync) models to curate a balanced 1,200-video benchmark dataset.
- Evaluated audio-visual synchronization as a deepfake detection signal, selecting a ~0.3 operating threshold through ROC analysis.
- Managed GPU-accelerated experimentation on shared SLURM clusters (NVIDIA H200 nodes) via Docker containers, running sanity-check protocols on custom validation splits before committing cluster time to full-scale benchmarking runs.
- Built React frontend components and FastAPI backend services with WebSockets for real-time streaming responses, deploying multi-model AI endpoints for an internal AI chatbot platform.

Research Assistant (Full Time / Volunteer), SUTD – ROAR Lab
Aug 2022 – May 2024

- Developed perception software processing 3D LiDAR data for autonomous mobile robots, designing architectures integrating Embedded C++, ROS, and sensor fusion.
- Led the design of the full intelligent lighting control system for Panthera, an autonomous vehicle developed under National Robotics Programme funding, implementing it as a rosserial node on an Arduino that subscribed to cmd_vel, IMU, and vehicle-state topics under an RTOS to drive turn, warning, and hazard signaling; the platform passed T1 autonomous testing at CETRAN.
- Built a predictive safety model to detect a stair-climbing robot's fall risk, deploying Long Short-Term Memory networks (LSTMs) over sensor time-series data.

Student Research Engineer, Singapore Institute of Technology (SIT)
Aug 2024 – May 2025

- Designed and built an experimental temperature chamber simulating human body heat signatures and extreme physical environments for faculty-led embodied AI and robotics hardware testing.
- Engineered a closed-loop, thermistor-based IoT control system to hold chamber temperature at setpoint, enabling high-fidelity data collection.
- Established repeatable testing and data collection protocols for faculty-led embodied AI and robotics hardware projects.

PROJECTS

1. AI-Generated Video & Audio Forensics Benchmarking
Sep 2025 – Aug 2026

- Reproduced and benchmarked state-of-the-art AI-generated video and audio detectors (NSG-VD, D3, ReStraV, Speech-Forensics) on SLURM-based HPC, achieving 90.03 AUROC on the NSG-VD replication.
- Integrated optical flow (RAFT), segmentation (SAM 2), and point-tracking (CoTracker) algorithms to analyze physical motion derivatives in video sequences for mission-critical data.
- Evaluated audio-visual synchronization as a deepfake detection signal, selecting a ~0.3 operating threshold through ROC analysis.

2. Synthetic Multimodal Deepfake Dataset Pipeline
Sep 2025 – Aug 2026

- Engineered a data generation pipeline chaining video extension (Wan2.2), voice cloning (Chatterbox), and lip-synchronization (LatentSync) models to curate a balanced 1,200-video benchmark dataset.

3. Multi-Model AI Endpoint Platform
Sep 2025 – Aug 2026

- Built React frontend components and FastAPI backend services with WebSockets for real-time streaming responses, deploying multi-model AI endpoints for an internal AI chatbot platform.
- Managed GPU-accelerated experimentation on shared SLURM clusters (NVIDIA H200 nodes) via Docker containers, running sanity-check protocols on custom validation splits before committing cluster time to full-scale benchmarking runs.

4. Stair-Climbing Robot Fall-Risk Prediction (LSTM)
Aug 2022 – May 2024

- Built a predictive safety model to detect a stair-climbing robot's fall risk, deploying Long Short-Term Memory networks (LSTMs) over sensor time-series data.

SKILLS / COMPETENCIES

AI, Vision & Audio Analytics: Deep Learning, Computer Vision, Multimodal ML, Deepfake Detection, Generative AI, OpenCV, Optical Flow (RAFT), Segmentation (SAM 2), Point-Tracking (CoTracker)
Machine Learning Frameworks: PyTorch, TensorFlow, Keras, Hugging Face Transformers, CNNs, DNNs, LSTMs, YOLO, Vision Transformers, Reinforcement Learning, Model Benchmarking, Zero-shot Learning
Software & Fullstack: Python (NumPy, Pandas), C++17, C#, Java, TypeScript, React, Zustand, FastAPI, OpenAPI, WebSockets, Git, Linux/Ubuntu, WSL
Infrastructure & Cloud Tech: SLURM, GPU Computing (CUDA), NVIDIA HPC Clusters (H200 nodes), Docker, Dev Containers, ONNX Runtime, uv, IoT Integrations, Agile/Scrum Methodologies

EDUCATION

DigiPen Institute of Technology & SIT
Aug 2023 – May 2027

- B.Eng. Mechatronics Systems
- Relevant Coursework: Mechanics, Control Systems, Robotics, Embedded Systems, Machine Learning, Systems and Software Engineering

Ngee Ann Polytechnic
Sep 2022 – Sep 2023

- Specialist Diploma in Robotics Engineering (SGInnovate PowerX)
- Relevant Coursework: Autonomous Robot Navigation, Machine Learning/Deep Learning for Robotics, Computer Vision

Singapore Polytechnic
Apr 2017 – Sep 2020

- Diploma in Mechatronics and Robotics
- Internship grade of B+; coursework in Microcontroller Applications, PLCs, and Robotics Integration

AWARDS / ACHIEVEMENTS / CERTIFICATES

- MOE (Ministry of Education) Instructor Certification (Achieved at age 18) — Valid Oct 2018 – Oct 2026
- Google Crash Course on Python
- Google The Bits and Bytes of Computer Networking
- Coursera Machine Learning